

AALBORG AP, Denmark

WMO: 060300

Lat: 57.096N

Lon: 9.851E

Elev: 3

StdP: 101.29

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -9.0	(c) -6.5	(d) -11.2	(e) 1.4	(f) -7.0	(g) -9.2	(h) 1.7	(i) -4.4	(j) 16.0	(k) 6.5	(l) 14.4	(m) 6.1	(n) 2.0	(o) 20	(p) 0.663

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.0	26.1	18.5	24.2	17.7	22.6	17.0	19.7	23.7	18.7	22.5	17.8	21.3	4.6	110

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.2	13.1	21.3	17.3	12.3	20.4	16.4	11.6	19.6	56.1	23.7	53.1	22.5	50.3	21.4	23.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 12.4	(b) 11.0	(c) 9.8	(e) -12.3	(f) 28.6	(g) 3.2	(h) 1.5	(i) -14.6	(j) 29.6	(k) -16.5	(l) 30.5	(m) -18.2	(n) 31.3	(o) -20.5	(p) 32.4		
(a) 12.4	(b) 11.0	(c) 9.8	(e) -12.7	(f) 21.3	(g) 3.1	(h) 0.9	(i) -14.9	(j) 22.0	(k) -16.7	(l) 22.6	(m) -18.4	(n) 23.1	(o) -20.6	(p) 23.8		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.7	1.0	1.2	3.3	7.4	11.3	14.6	17.2	17.1	13.6	9.4	5.3	2.3
	DBStd	6.66	3.81	3.69	3.44	3.17	3.15	2.64	2.64	2.42	2.51	2.99	3.27	4.21
	HDD10.0	1277	279	246	209	89	21	1	0	0	2	47	143	240
	HDD18.3	3566	538	479	468	327	218	118	54	54	142	278	391	498
	CDD10.0	795	0	0	1	12	62	137	224	219	111	27	2	0
	CDD18.3	41	0	0	0	0	1	4	20	15	1	0	0	0
	CDH23.3	256	0	0	0	0	9	40	134	71	2	0	0	0
	CDH26.7	29	0	0	0	0	0	5	20	4	0	0	0	0
Wind	WSAvg	4.8	5.3	5.3	5.2	4.7	4.5	4.7	4.3	4.3	4.6	4.8	4.8	4.8
Precipitation	PrecAvg	622	46	31	35	35	45	59	63	68	66	66	60	51
	PrecMax	875	102	88	87	81	120	124	200	202	177	153	120	110
	PrecMin	431	4	2	1	5	5	13	6	11	21	15	14	9
	PrecStd	107	23	19	19	18	24	32	40	39	31	32	25	25
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.2	9.4	13.3	19.6	24.8	27.2	29.0	27.2	23.2	17.8	13.7	11.0
		MCWB	8.1	7.8	9.0	13.4	17.0	18.1	19.0	18.6	17.8	14.7	12.4	10.0
	2%	DB	8.0	8.0	10.8	17.2	21.8	24.2	26.6	25.2	21.0	15.8	12.1	9.5
		MCWB	6.9	7.0	7.7	11.4	15.2	17.0	18.9	18.3	16.7	13.8	11.0	8.7
	5%	DB	7.0	7.0	9.1	14.9	19.3	21.8	24.8	23.7	19.2	14.6	10.9	8.6
		MCWB	6.0	6.0	6.9	10.5	14.0	16.0	18.2	17.8	15.7	13.0	9.9	7.7
	10%	DB	6.1	6.1	7.9	12.8	17.2	19.7	22.7	21.9	17.8	13.5	9.9	7.6
		MCWB	5.2	5.2	6.2	9.3	12.9	15.1	17.2	17.3	14.9	12.1	8.9	6.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	8.3	8.2	9.8	14.2	18.0	19.7	21.0	20.8	18.7	15.6	12.6	10.1
		MCDB	8.9	8.9	12.2	18.2	22.9	24.3	26.0	24.4	21.4	17.0	13.4	10.7
	2%	WB	7.1	7.1	8.3	12.2	16.1	18.1	19.8	19.6	17.4	14.2	11.2	8.8
		MCDB	7.7	7.7	10.0	15.9	20.3	22.5	24.5	23.0	20.2	15.4	12.0	9.3
	5%	WB	6.1	6.1	7.2	10.8	14.6	16.8	18.9	18.6	16.3	13.2	10.0	7.8
		MCDB	6.7	6.7	8.6	14.0	18.4	20.6	23.1	22.0	18.5	14.3	10.7	8.4
	10%	WB	5.3	5.3	6.3	9.7	13.3	15.6	17.9	17.7	15.4	12.4	8.9	6.8
		MCDB	5.9	5.9	7.5	12.3	16.5	18.8	21.6	20.8	17.3	13.5	9.7	7.4
Mean Daily Temperature Range	5% DB	MDBR	4.2	4.5	5.9	7.7	8.2	7.6	8.0	7.6	6.7	5.5	4.4	4.2
		MCDBR	4.2	4.6	7.4	10.9	11.7	11.7	11.4	10.4	8.7	6.0	4.4	4.1
		MCWBR	4.1	4.1	5.1	6.4	6.5	6.1	5.3	5.1	5.4	4.5	4.1	4.1
	5% WB	MCDBR	4.1	4.3	6.6	9.8	10.1	9.9	9.8	8.6	7.7	5.5	4.2	4.1
MCWBR		4.1	4.0	4.9	6.1	6.2	5.9	5.3	4.9	5.2	4.5	4.0	4.1	
Clear-Sky Solar Irradiance	taub	0.308	0.333	0.351	0.388	0.380	0.371	0.390	0.392	0.365	0.347	0.324	0.294	
	taud	2.224	2.285	2.322	2.274	2.351	2.400	2.367	2.371	2.431	2.400	2.289	2.198	
	Ebn at Noon	551	707	800	823	856	869	841	813	781	688	531	456	
	Edh at Noon	53	75	93	114	112	109	110	103	84	67	50	41	
All-Sky Solar Radiation	RadAvg	0.39	0.94	2.36	4.03	5.36	5.89	5.42	4.24	2.85	1.37	0.51	0.27	
	RadStd	0.08	0.17	0.21	0.46	0.48	0.45	0.52	0.30	0.28	0.17	0.09	0.03	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		+0.90	N/A	N/A	N/A	N/A	N/A	-200	-331	+122	N/A
Regional (46 neighbors)		+0.35	N/A	N/A	N/A	N/A	N/A	-119	-156	N/A	N/A

Nomenclature: See separate page